

Legislative Coordination and Institutional Change

A Bipartite Network Analysis of Coalition Dynamics in the Brazilian Chamber of Deputies

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| The puzzle

- ◆ Polarization should make legislatures **ungovernable**.
- ◆ Brazil 2014–16: impeachment · corruption scandals · recession · polarization.
- ◆ Yet the Chamber **kept forming majorities**, across impeachment and a change of power.
- ◆ The literature's crisis typology: **collapse · fragmentation · realignment · institutionalization**. Brazil fits **none**.
- ◆ **How? And at what cost?**

Coalitional presidentialism, and a cluster of shocks

- ◆ Strong president, **no majority** → govern by **coalition** (Abranches 1988).
- ◆ Levers: **budget · posts · agenda**; the **Centrão** sells support as the president's *agent*.
- ◆ Recent work reads a **weakening presidency and a rising Congress**, much of it via **budget powers** once a pillar of the president's toolkit.
- ◆ 2014–16, all at once: impeachment · corruption scandals · recession · **and the president's budget lever removed**.
- ◆ **Shocks co-occur in a tight 2014–16 window** → no single lever separates cleanly from the rest (hence the eliminative design).

| The structural question, and some rival explanations

- ◆ The open question is structural: **How did coordination reorganize after the 2014–2016 shocks? What happened to the pragmatic center, and at what cost?**
- ◆ Then rule out the rival explanations, one at a time:
- ◆ Is it **turnover** (new members) or a one-off **crystallization**?
- ◆ Is it the **Cunha effect** (the agenda-setter) or the **fiscal channel**: does re-sorting concentrate on economic-fiscal rather than rights-and-moral bills?

The tests ask whether a durable pivot remains once turnover, agenda, and fiscal explanations are removed.

| Why networks, not ideal points

- ◆ **deputies × bills** → agreement graph → **communities**.
- ◆ Standard **one-dimensional ideal-point** models compress change onto one axis.
- ◆ Blocs **recompose** while ideology looks stable; an axis can't see it.
- ◆ No axial assumption → the shift is **bloc recomposition**.

| Related work, and how this differs

- ◆ **Prior networks of the Chamber exist** (Brito 2020; Toledo Jr. 2025). **Four differences:**
- ◆ **1. Bipartite primitive:** starts from the deputy×bill incidence and explicitly tests projection dependence.
- ◆ **2. Null-corrected structure:** modularity against a **within-vote permutation null (Q^*)**, not a raw score.
- ◆ **3. Within-legislator design:** separates a change in behavior from a change in composition.
- ◆ **4. Theory-driven object:** the object is **which coordinating community a bloc occupies relative to the executive**, not a left–right axis.

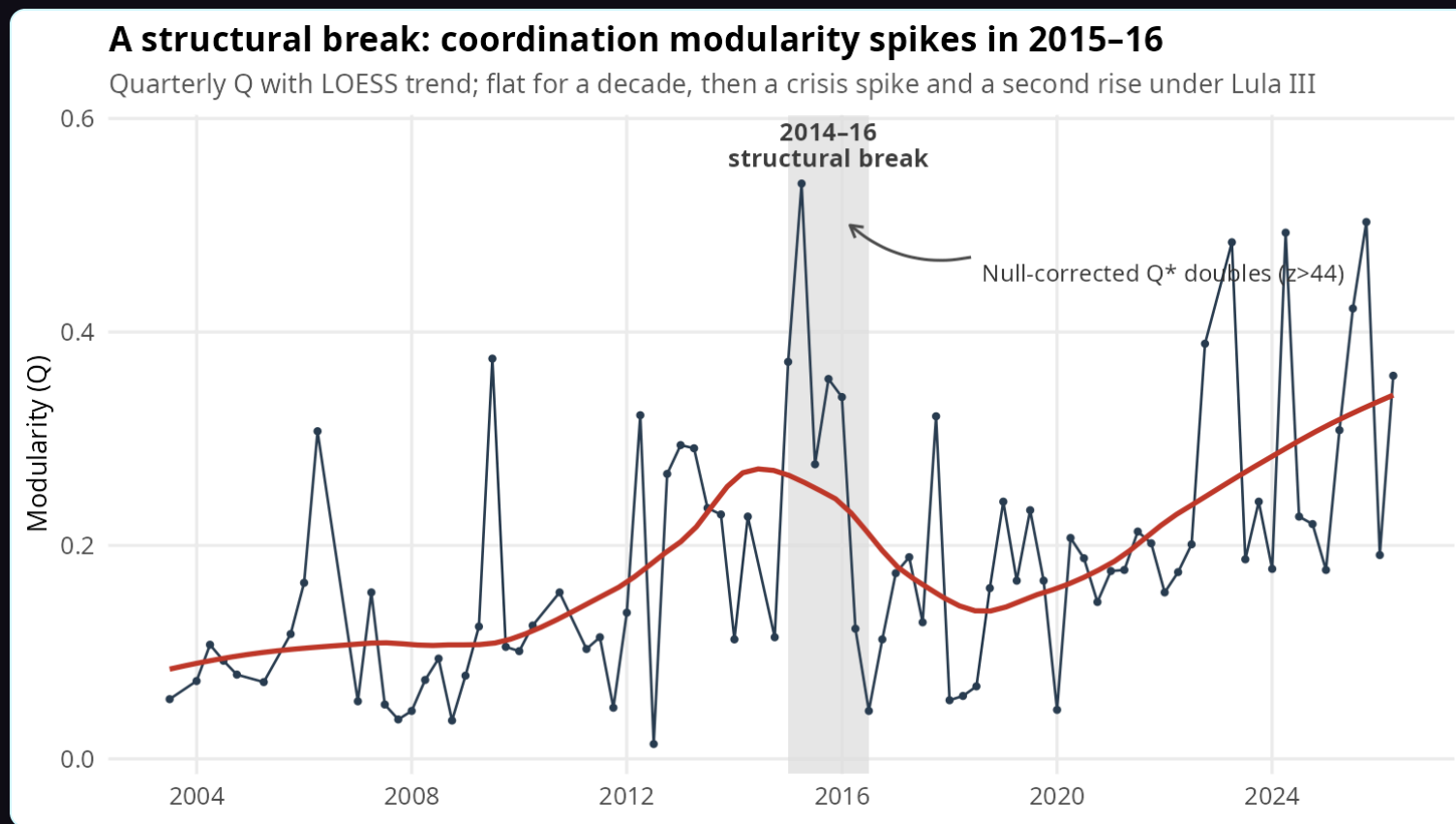
| The methodological approach

- ◆ **Measure:** roll-calls → **deputy × bill** graph → **communities per legislature**.
- ◆ **Classify bills:** fiscal vs moral, **rule-based** + local **LLM cross-check** (Qwen 2.5 32B), human-validated (n=294, $\alpha=0.84$, macro-F1 0.88, 89.5%).
- ◆ **Compare:** across the 2014–16 break; a **within-deputy panel** separates behavior from turnover.
- ◆ **Falsify, not estimate:** near-collinear shocks → **eliminate rivals one at a time**.
- ◆ **Stress-test:** Q^* null, 12-cell threshold grid, alternative projections and algorithms.

| What the ERGM exercises establish

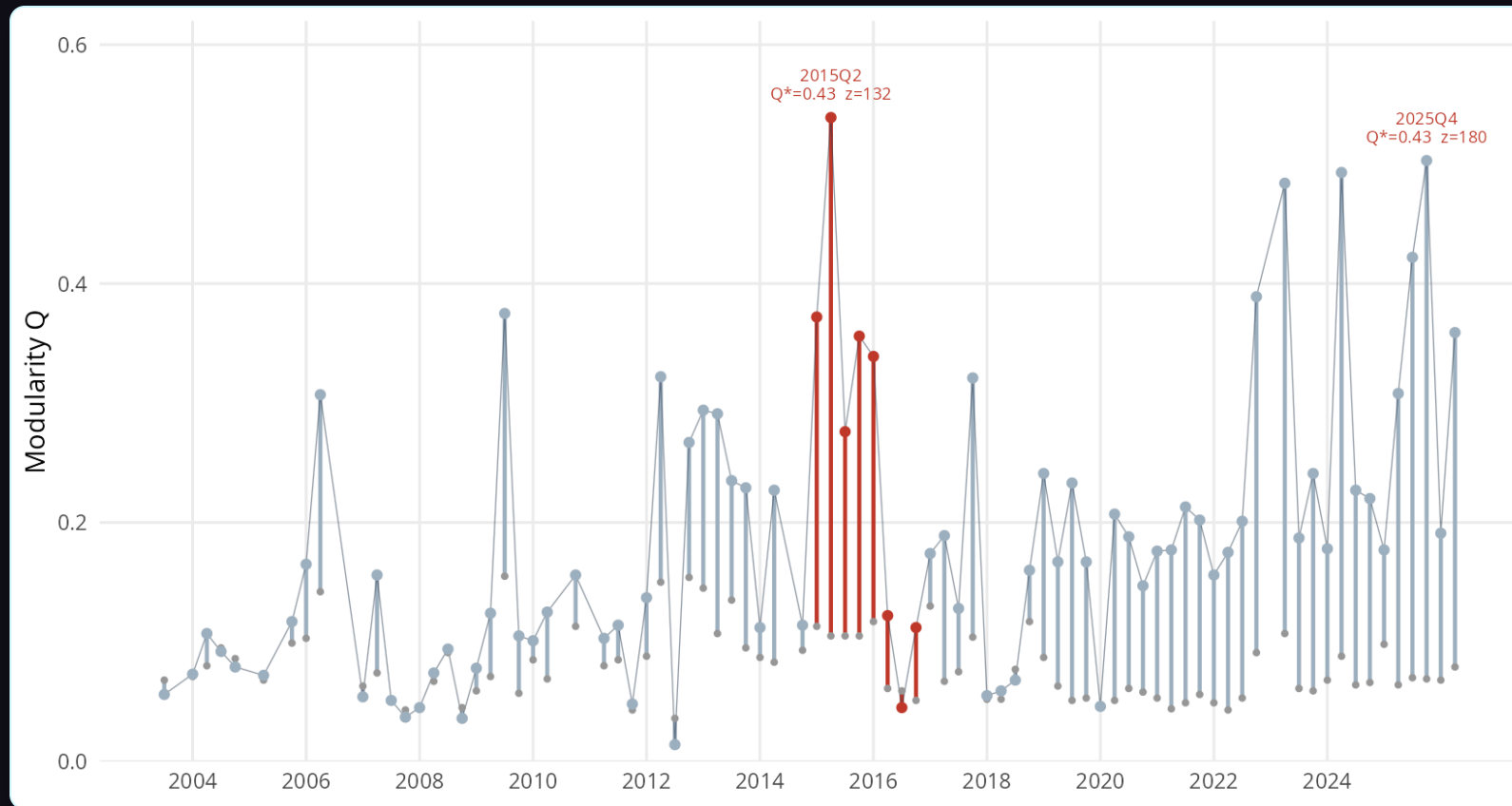
- ◆ Estimand: the **causal effect of mandatory amendments** (EC 86).
- ◆ **Static bipartite ERGM** (deputy×bill incidence): **did not mix** on the dense 54L network.
- ◆ **Temporal agreement-network model** (btergm): **runs**, recovering **closure** and **tie persistence**.
- ◆ **Identification remains**: neither exercise separates EC 86 from the post-2014 shock cluster (VIF 11.6).
- ◆ **Implication**: estimation can *model* coordination, but cannot *isolate* the reform's marginal effect.

A structural break: modularity spikes in 2015–16



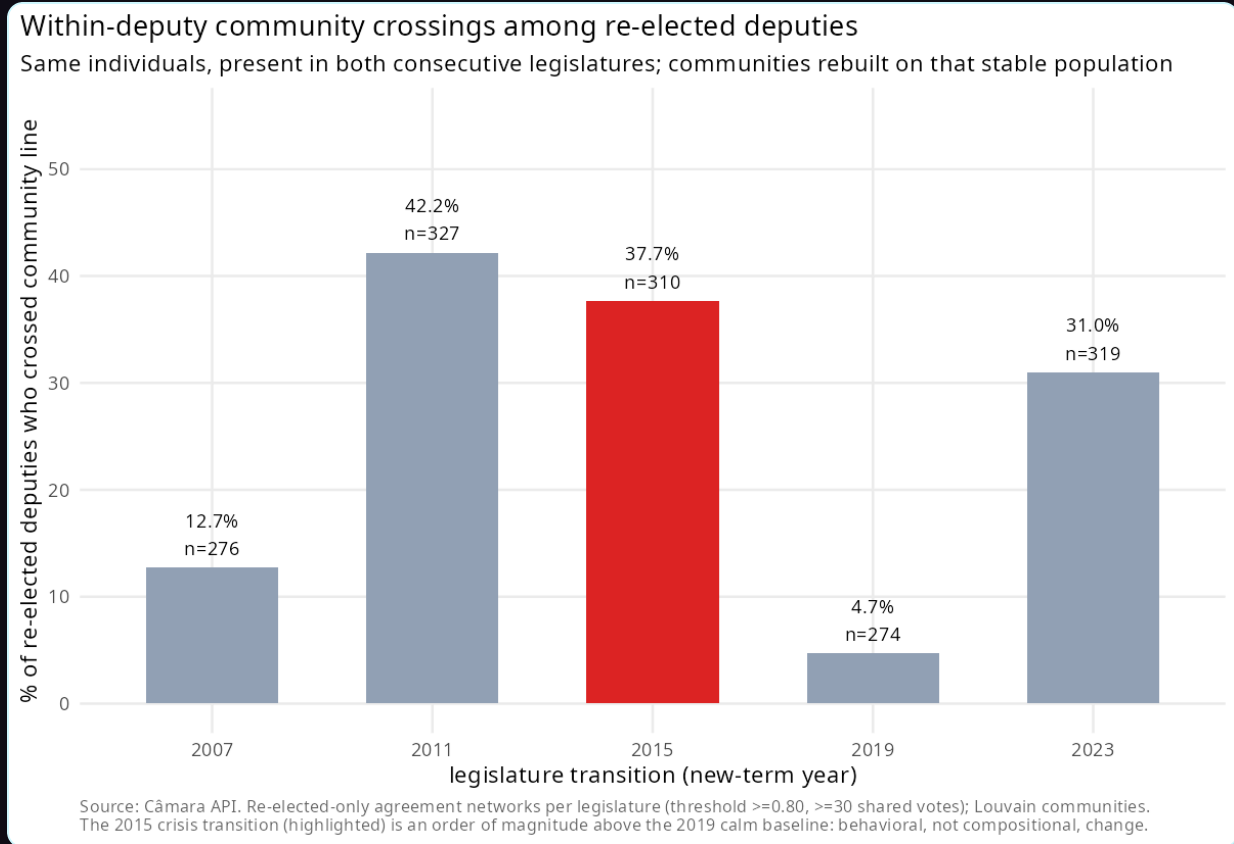
Quarterly modularity rises from $\approx .15$ to $.51$ in 2016.

Null-corrected modularity (Q^*) over time



Observed Q exceeds the within-vote null, peaking in 2015–16 and rising again under Lula III.

Same people, not turnover

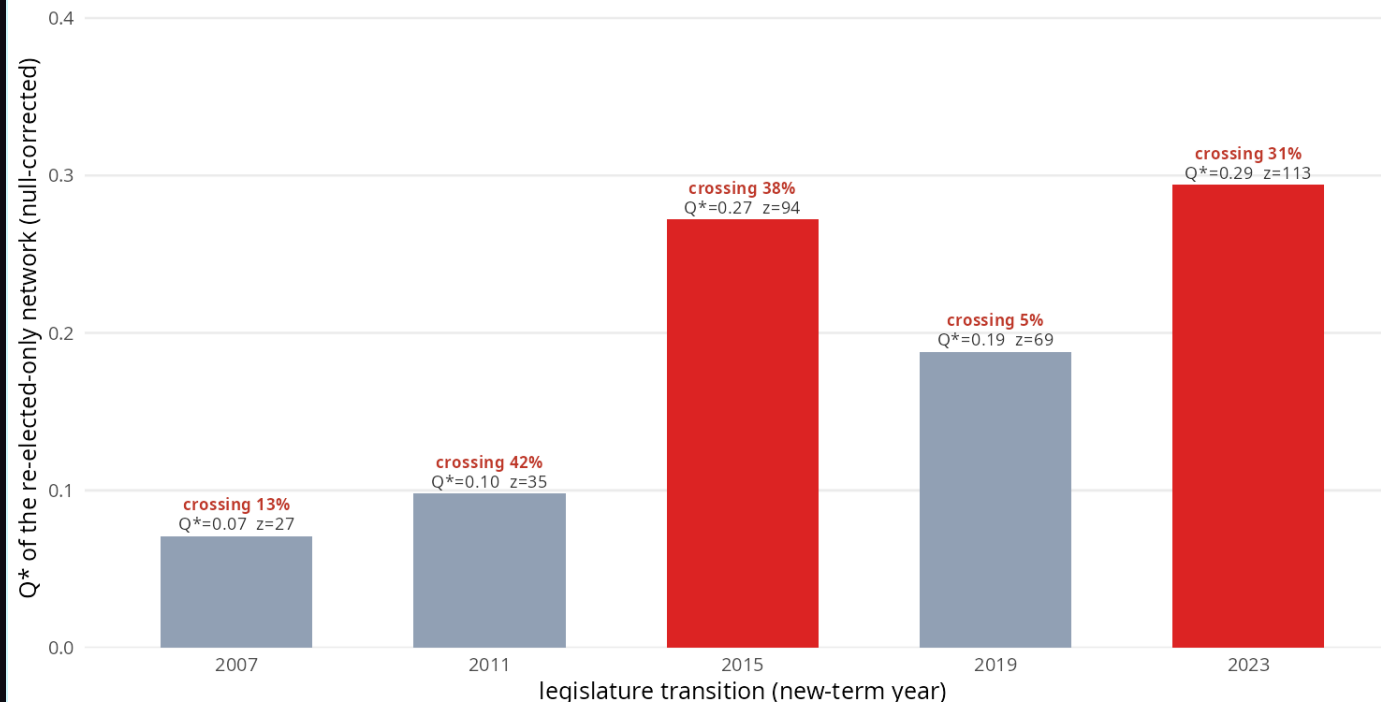


Reelected-deputy crossing: **37.7%** at the break, **4.7%** afterward.

Movement is not crystallization

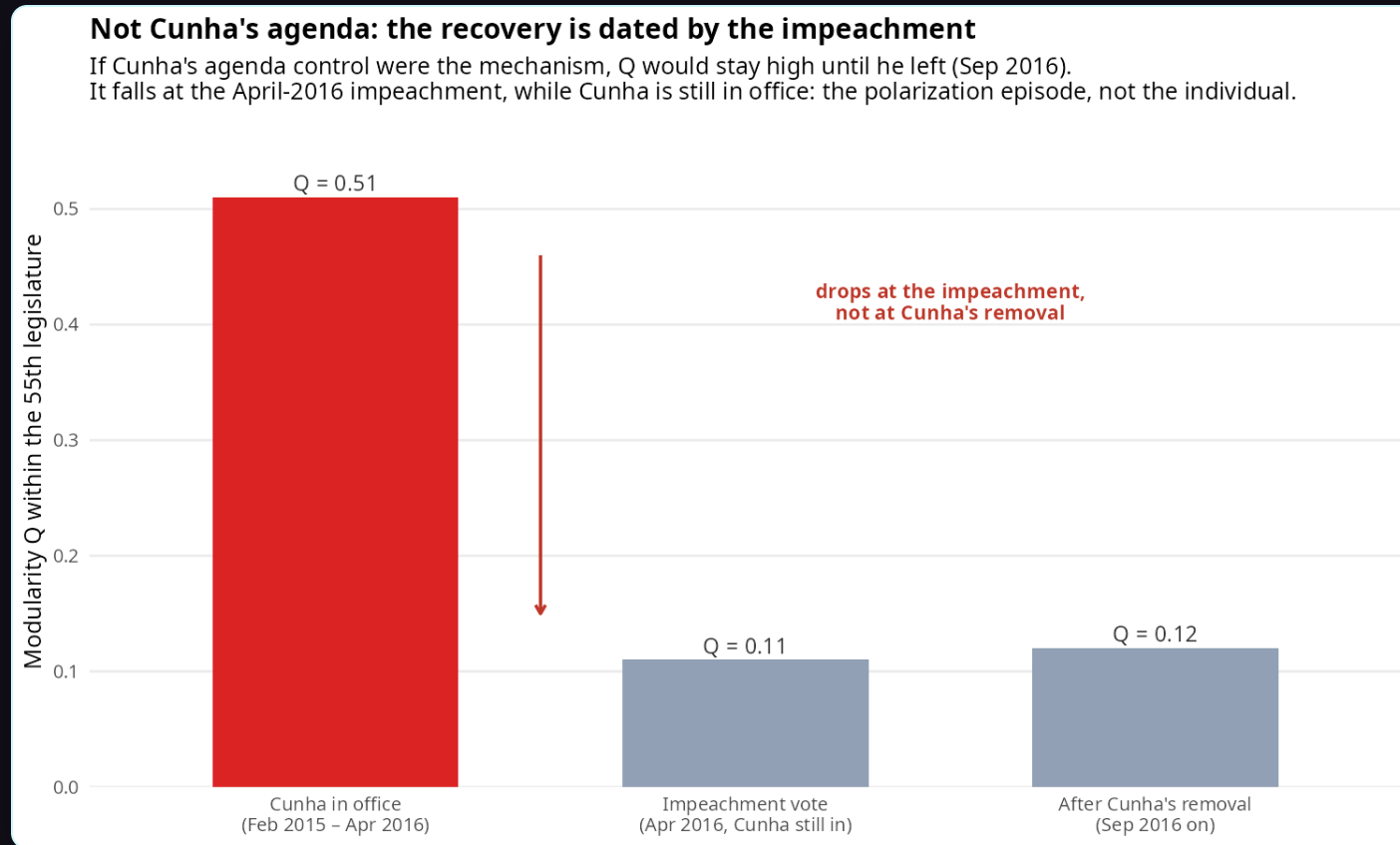
Same people: movement vs crystallization

Null-corrected modularity among deputies present in both legislatures. Crossing is movement, Q^* is structure. The two re-sorts (2015 crisis, 2023 Lula III alternation) move the same people AND sharpen the structure; 2011 churns (high crossing, low Q^*) and 2019 holds (consolidated, little movement).



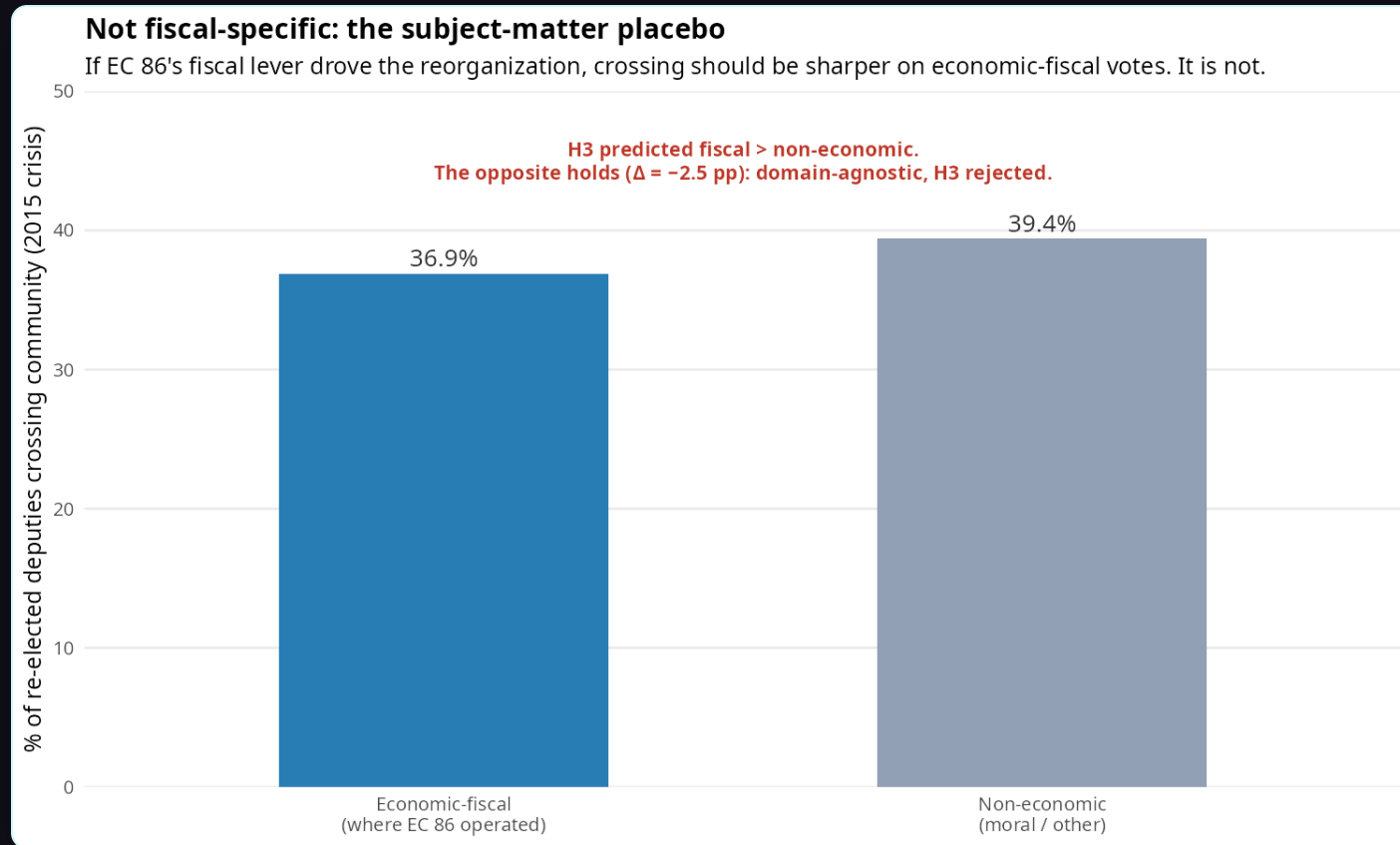
2011 moves more deputies but stays diffuse; 2015 and 2023 sharpen the structure.

Not Cunha's agenda: dated by the impeachment



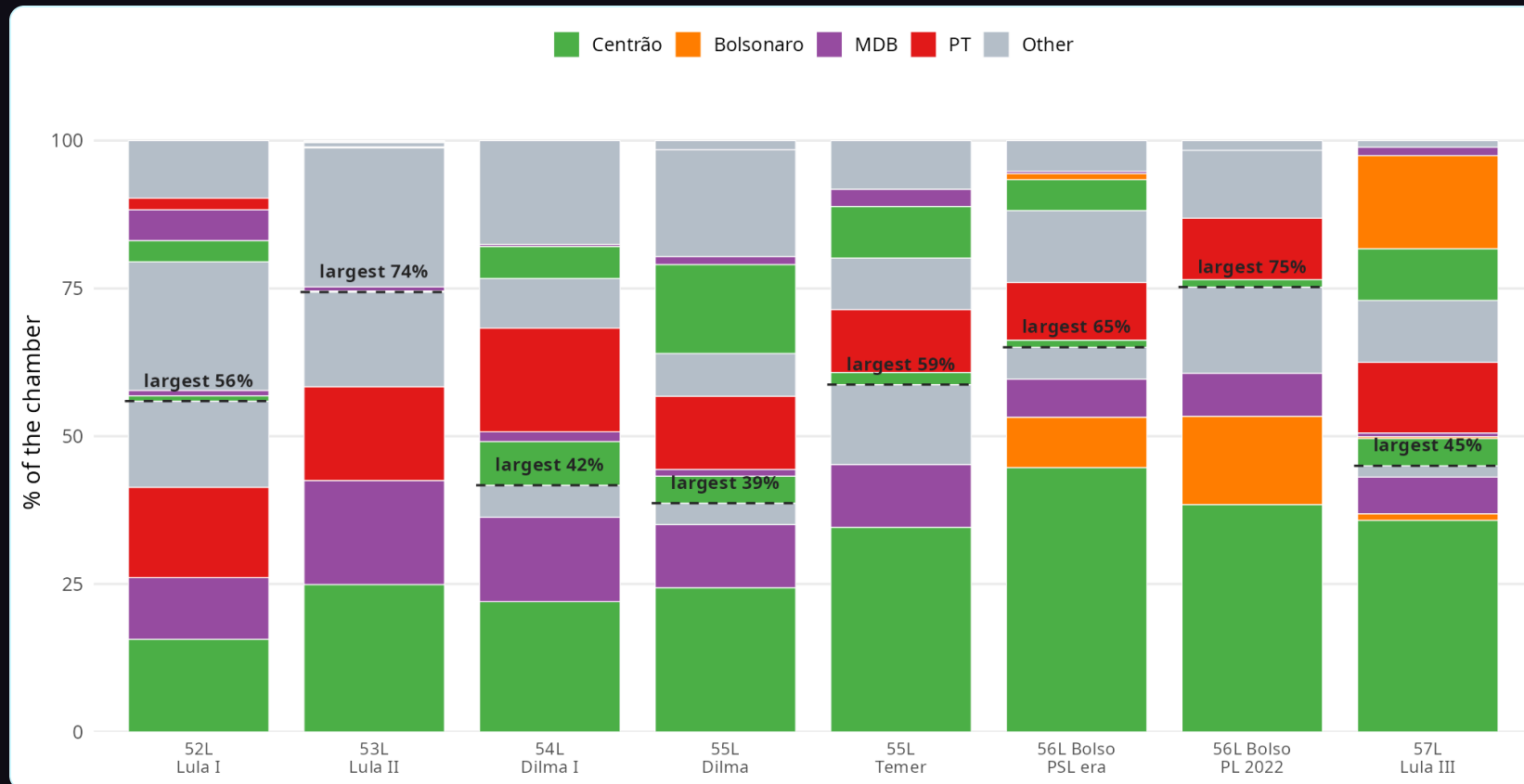
Modularity falls at the impeachment, not at Cunha's removal.

Not fiscal-specific: the subject-matter placebo



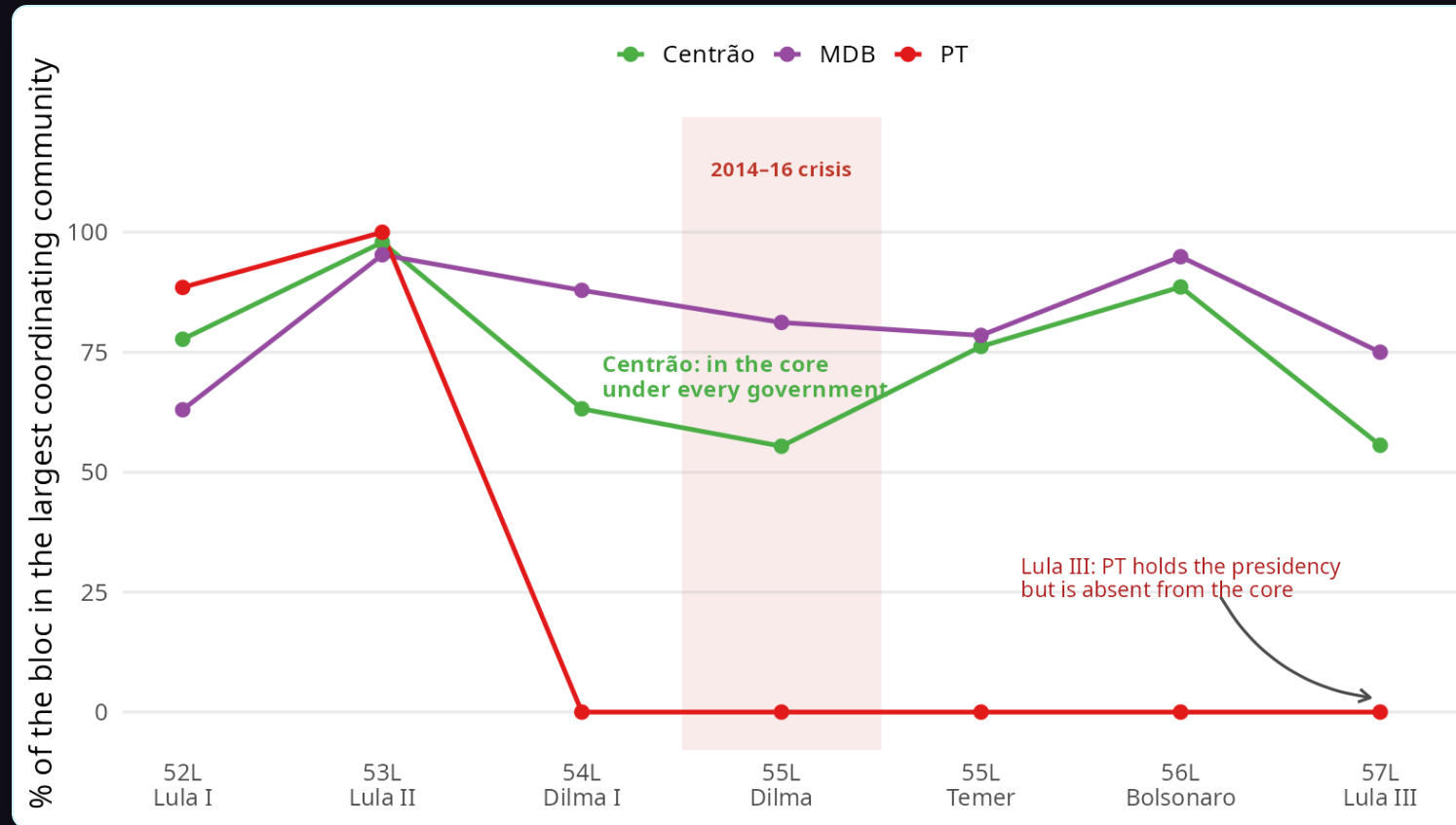
Fiscal: 36.9%; non-economic: 39.4%; $\Delta = -2.5$ pp.

The coordinating communities, sized



From 2014, the Centrão anchors the largest community while the PT moves to a smaller one.

The pivot: in the largest community under every government

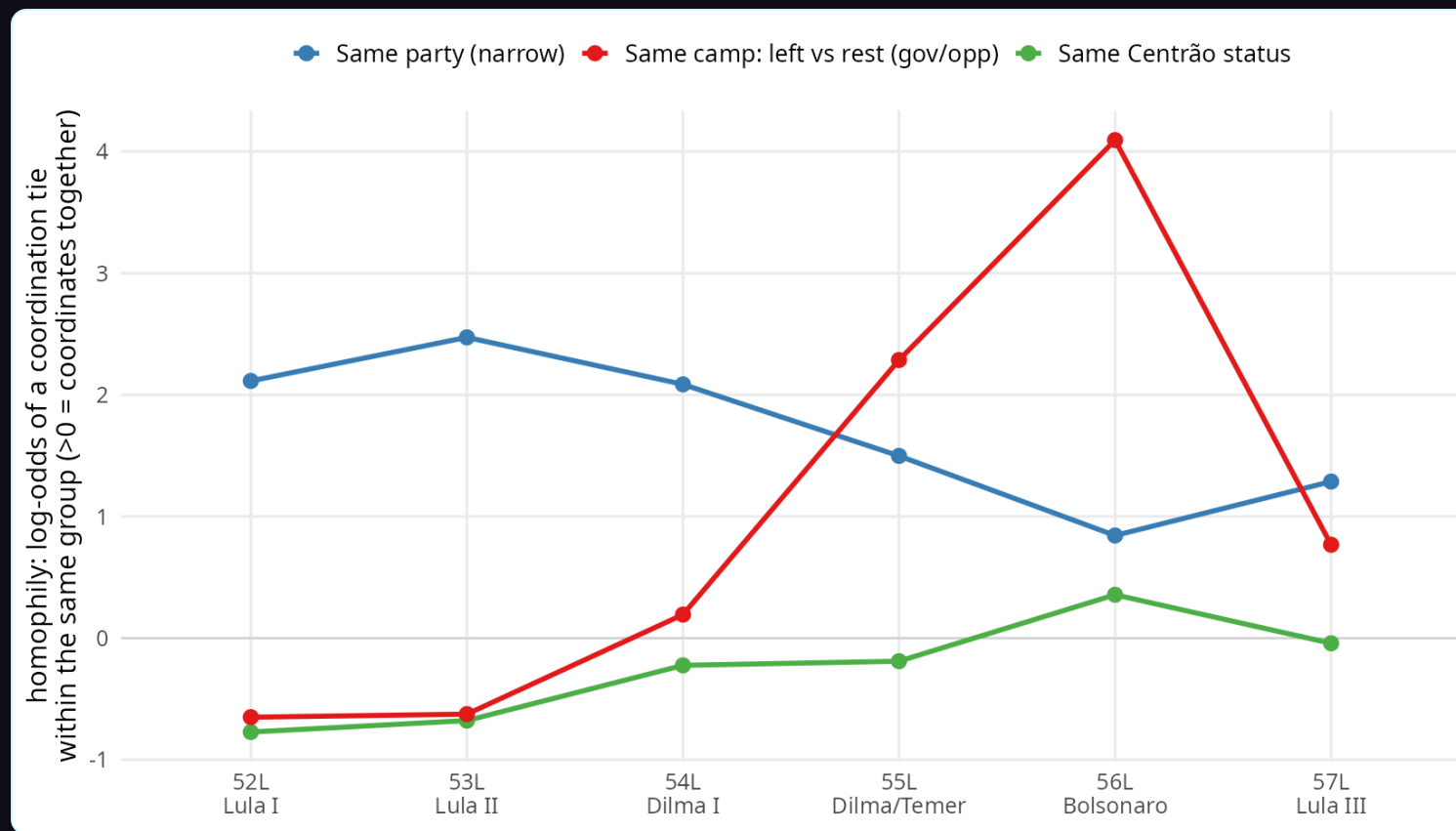


The Centrão remains in the largest community across governments; the PT does not.

| Can we trust these communities?

- ◆ **Noise?** Q^* sits far above a within-vote permutation null ($z > 44$).
- ◆ **Projection?** A projection-free bipartite SBM shows compatible alignment at the 56L (ARI 0.23), weakening the projection-artifact interpretation.
- ◆ **Labels?** Vote-only communities increasingly align with an independently coded Centrão (ARI 0 $\rightarrow$ 0.39).
- ◆ **Model dependence?** A descriptive ERGM detects the same shift, from party homophily to government–opposition organization.
- ◆ **Just the PT in opposition?** No: the government side organizes as strongly as the opposition, and the shift survives dropping the PT.

What organizes coordination: party gives way to position



Party homophily declines as government-opposition organization rises after 2014.

| The thesis, and what it means

- ◆ The pragmatic center has become the recurring pivot through which governments build majorities.
- ◆ Coordination by **position relative to the executive**, beyond a fixed ideological axis.
- ◆ The Centrão is **old** (1987); its **position** changed: *agent* → *pivot*.
- ◆ What it means: governability **sold, not represented**; the cost is **accountability**.

| From agent to principal, and the cost

- ◆ **Why the position flips (a reading, not a tested cause):** the 2014–16 rupture began the transfer to Congress (**mandatory amendments**); later instruments, the **secret budget** and **strategic agenda control**, consolidated it, so the tool that once disciplined the pivot now lets it **set the terms** (*governo congressuale*).
- ◆ Governability comes to be **sold, not represented**.
- ◆ Cost = **accountability**, not gridlock: a bloc available to any government is hard to hold to account.

| Contribution, and what the case teaches

- ◆ **Measurement:** recover the Centrão's permanence from the votes, its position and persistence, where the literature had narrative.
- ◆ **Design:** with near-collinear shocks, eliminate the rival explanations rather than estimate one effect.
- ◆ **A portable template:** turn a narrative prediction, that institutional crisis reshapes the legislature, into a measurable, falsifiable object, and narrow the viable mechanisms.
- ◆ **A way of looking, not a new claim:** the permanent-pivot type is old (Israel, post-war Italy).

Thank you

Questions welcome.

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